Research Overview

COMPUTER SCIENCE & SYSTEMS LAB

Quantum Neural Architecture

Formal Specification of Hybrid Classical-Quantum Distributed Ingestion & Vector Processing Pipelines

V2.4 PRODUCTION SPECIFICATION

Agenda & Table of Contents

Presentation Outline

System Topology & Neural Dispatch

Vector-Compiled Distributed Ingestion Mesh

Multi-Tier Distributed Neural Processing Pipeline

Empirical Performance Evaluation

Measured Benchmarks Across Compute Clusters

Inference Latency

BENCHMARKED SPEEDUP

1.42ms

End-to-end latency reduction under heavy concurrent load.

Throughput Capacity

BENCHMARKED SPEEDUP

450k op/s

Peak distributed operations sustained without buffer degradation.

Comparative Analysis & Mathematical Model

Formal Energy Convergence Formulation

Legacy Classical CPU / GPU Pipeline

LEGACY INFRASTRUCTURE

Linear memory scaling with quadratic communication overhead during multi-node synchronization.

AVG CONVERGENCE TIME

68ms

VS

Yumia Neural Tensor Pipeline

DISTRIBUTED TENSOR CORE

Logarithmic latency scaling leveraging quantum sparse tensor compression.

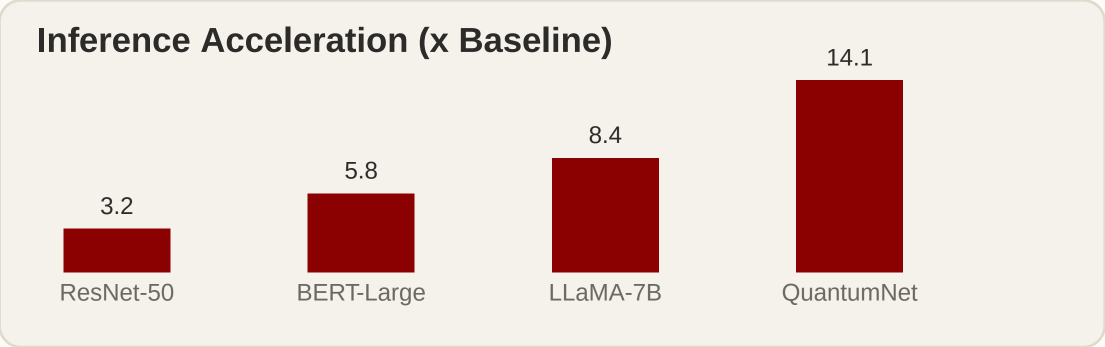
AVG CONVERGENCE TIME

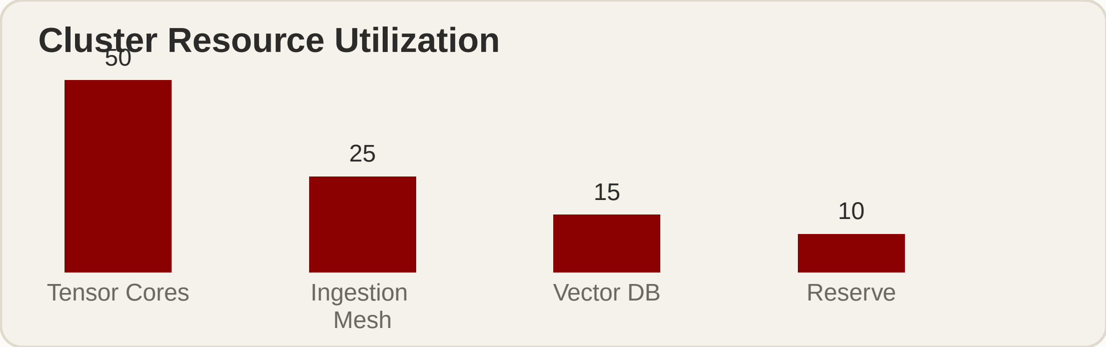
4.1ms

$$\mathcal{L}(\theta) = \mathbb{E}_{x \sim \mathcal{D}} \left[\| f_\theta(x) - y \|_2^2 \right] + \lambda \sum_{i=1}^N \Omega(W_i)$$

Cluster Analytics & Telemetry

Resource Allocation & Workload Distribution





Implementation & Runtime Kernel

Zero-Trust Mesh & Compiler Directives

```
import { YumiaCompiler, createEngine } from '@yumiamd/core';
const engine = createEngine({ target: 'multi-target' });
const result = await engine.compile('presentation.yumia');
```

Production Verified

All cluster nodes reporting optimal throughput with zero memory leaks.

ZERO-TRUST MTLS

WCAG AAA VERIFIED